AIIQ

An AI-Powered, Source-Grounded Learning & Assessment Platform

AIIQ converts a student's own lecture notes, textbooks and slides into quizzes, flashcards, study notes and a cited Q&A assistant — then measures mastery topic by topic to direct revision where it is actually needed.

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PROGRAMME
MSc Data Science

DOMAIN
Applied NLP & Learning Analytics

LEARN IT. QUIZ IT. WIN IT.

01 Project Overview

AIIQ is a full-stack learning platform built around a single constraint: **every item it generates must be answerable from the material the student uploaded** — never from general knowledge.

A document is parsed to text, sent to a large language model under a strict grounding contract, and returned as structured JSON. From one upload the system produces four distinct study artefacts.

Every attempt is recorded and analysed, so the platform closes the loop from **source material → assessment → measured weakness → targeted revision**.

02 Problem

- **Generic AI output.** General-purpose assistants produce plausible but off-syllabus questions, so practice does not reflect the actual course.
- **Unverifiable answers.** Students cannot tell which parts of an AI answer came from their material and which were invented.
- **Undirected revision.** A single percentage score says nothing about *which* topics are weak, so learners re-read what they already know.
- **Fragmented tooling.** Quiz generators, flashcard apps and note summarisers are separate products that never share progress data.

03 Objectives

01 Generate assessment **strictly from user-supplied sources**, with outside knowledge explicitly prohibited at the prompt level.

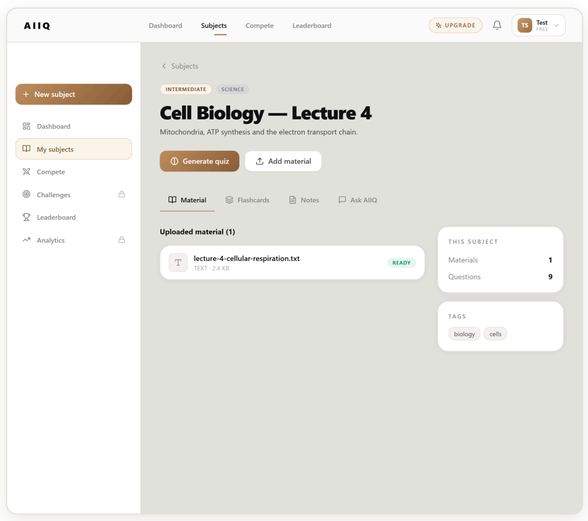
02 Support **four question formats** across four difficulty bands.

03 Measure **mastery per topic** rather than a single aggregate score.

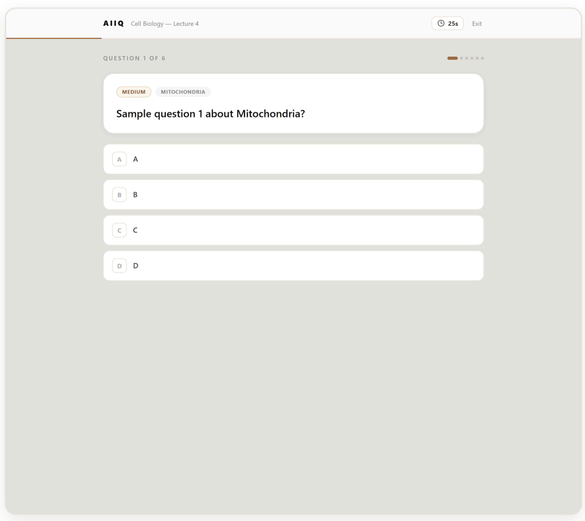
04 Automatically **direct revision** to the weakest topics after every attempt.

05 Deliver one **connected journey** — upload, generate, assess, analyse — inside a single application.

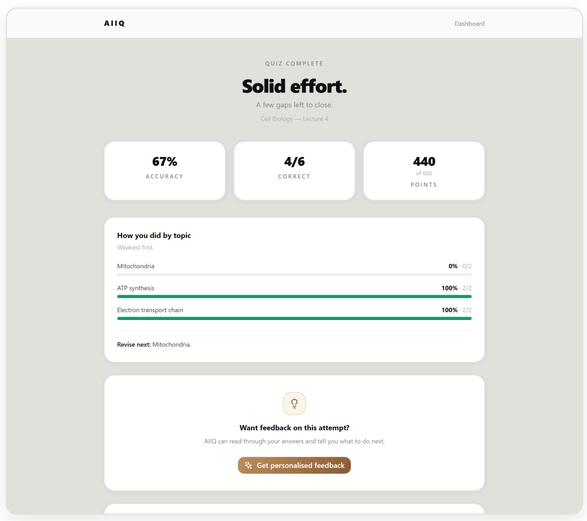
04 The Application



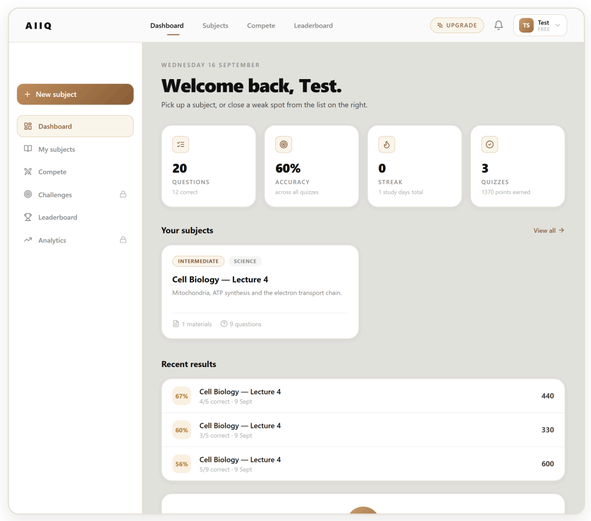
Subject workspace
One upload drives quizzes, flashcards, notes and Q&A.



Quiz runner
Timed and distraction-free; resumes after a refresh.



Results
Score, per-topic accuracy and revision guidance.



Dashboard
Derived mastery stats and weakest-topic ranking.

05 Key Features

AI quiz generation
MCQ, true/false, fill-in-the-blank and short answer, from Easy to Expert.

Flashcard decks
Term/definition cards with shuffle and know-or-review tracking.

Study notes
Overview, key points, concept map and study tips per subject.

Ask AIIQ
Streaming Q&A that answers only from uploads and cites the source line.

Timed quiz runner
Per-question countdown, keyboard answering, resumes after refresh.

Adaptive coaching
Post-attempt feedback naming weak topics and the next actions.

Topic analytics
Accuracy per topic plus a 14-day activity history.

Live quiz rooms
Real-time multiplayer over WebSockets with ELO and leaderboards.

Result export
Full attempt data downloadable as CSV or JSON.

06 Technology

FRONTEND	Next.js 15 (App Router), React 18, TypeScript, Tailwind CSS
BACKEND	Next.js route handlers, custom Node server, Zod validation
DATABASE	PostgreSQL with Prisma ORM — 28 models, 11 enums
REALTIME	Socket.IO server for live quiz rooms
AUTH	JWT (jose) + bcrypt, httpOnly cookies, route middleware
PAYMENTS	Stripe subscriptions (three tiers) with webhooks
INFRA	Docker Compose — PostgreSQL and Redis

23	27	28	13.4k
PAGES	API ROUTES	DB MODELS	LINES TS

07 AI & Data Science Components

01 Extract — PDF, DOCX, PPTX, TXT and image OCR → **02 Ground** — Strict source-only prompt contract → **03 Generate** — JSON-constrained LLM output → **04 Analyse** — Derived per-topic mastery

MODEL	Google Gemini 2.5 Flash — JSON-constrained, vision-capable for image OCR
GROUNDING	Prompt contract forbids outside knowledge; produces fewer items rather than padding
DEDUP	SHA-256 fingerprint of normalised question text (16-hex prefix) blocks repeats
ANALYTICS	Accuracy, streaks and topic mastery derived from attempt records, never stored counters

$$score = 100 \times (0.5 + 0.5 \times t_{remaining} / t_{total}) \times d$$

Time-weighted scoring. Difficulty multiplier d = 1.0 Easy · 1.5 Medium · 2.0 Hard · 3.0 Expert. Speed is rewarded but accuracy dominates — a wrong answer scores zero regardless of time.

08 Results & Evaluation

Method. A single 377-word cellular-respiration lecture was uploaded as the only source. Every artefact the system produced was checked against that text, and the analytics layer was exercised with attempt data carrying known expected values.

A · GENERATION FROM ONE UPLOAD	
Flashcards produced, each tagged to a topic present in the source	**16 cards**
Structured study notes — overview, key points, concept map, study tips	**8 key points**
Duplicate questions suppressed by SHA-256 fingerprint before storage	**0 repeats**
B · GROUNDING FIDELITY	
Q&A answers quoted the originating sentence from the upload verbatim	**Cited**
No outside knowledge appeared in any artefact; thin input yields fewer items rather than invented padding	**On-source**
C · MEASUREMENT ACCURACY	
Derived accuracy reproduced the expected value exactly (8 of 14 correct)	**57% · exact**
Per-topic mastery aggregated correctly across multiple attempts — electron transport 1/4, ATP synthesis 3/5, mitochondria 4/5	**25 / 60 / 80%**
Weakest-topic ranking drove the revision recommendation surfaced to the learner	**Correct order**
D · ADAPTIVE FEEDBACK	
Coaching identified the failed topic (0 of 2 on mitochondria) and named both mastered topics	**Identified**
Recommended revision questions matched precisely the items answered incorrectly	**2 of 2**

09 Future Scope

Adaptive difficulty
Activate the item-response-theory field already modelled in the schema to tune question difficulty per learner.

Spaced repetition
Schedule flashcard review intervals from per-card recall history.

Horizontal scaling
Move live-room state from memory to Redis so realtime runs multi-instance.

Battle validation
Full multi-player load testing, with ELO driven by live match outcomes.

Institution mode
Cohort dashboards so tutors can see aggregate topic weakness across a class.